Saba Azadegan

Structural engineer (MSc) · AI trainer for civil and structural engineering
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PROFILE

Structural engineer with an MSc in structural engineering and site experience on a cast-in-place reinforced concrete project, where I carried out the full quantity takeoff and cost estimate and supervised foundation and reinforcement work. My academic work covered structural design and structural dynamics, with building analysis in ETABS and SAFE and finite element modelling in Abaqus. I now design expert-level structural engineering tasks and answer keys that test how well AI models apply reinforced concrete design and US building codes. I also hold an MSc in Business from Católica Lisbon.

ENGINEERING EXPERIENCE

AI Trainer, Civil and Structural Engineering

DataAnnotation · Remote, freelance · Oct 2026 to present

- Design expert-level structural engineering tasks, with realistic drawing sets, specifications and answer keys, that expose where AI agents misapply reinforced concrete design rules.
- Check every expected answer by hand against the code text, mainly ACI 318-19 and ASCE 7-16, and set fair tolerances for alternative methods a competent engineer could use.
- Review AI output for errors with real safety or cost consequences and document clearly why each answer is right or wrong.

Civil Engineer (internship), Parvin mixed-use project

Aghigh Naghsh-e-Jahan Investment Holding · Isfahan, Iran · May 2018 to Oct 2018

Commercial, office and multi-storey parking building in cast-in-place reinforced concrete.

- Carried out the full quantity takeoff and cost estimate for the structure: concrete volumes and rebar quantities for foundations, columns, beams and slabs, taken from AutoCAD structural drawings and calculated in Excel.
- Supervised foundation works and rebar mesh installation on site.
- Attended concrete pours and visited the ready-mix supplier's laboratory.

EDUCATION

MSc in Civil Engineering, Structural Engineering

Kharazmi University, Iran · Sep 2018 to Feb 2021 · GPA 15.75/20 (programme average 14.95)

- About a year and a half of work in structural dynamic analysis, including damage detection in bridges.
- Finite element modelling in Abaqus; top grade of 18/20 in Finite Element Methods.

BSc in Civil Engineering

Azad University, West Tehran Branch · Sep 2012 to Dec 2016 · GPA 15.84/20

- About two years of structural design projects, with analysis and design in ETABS and SAFE.
- Spaghetti Structures competition, Shahid Beheshti University, 2015.
- Professional training: AutoCAD (2015) and ETABS and SAFE (2016), Tehran Institute of Technology.

MSc in Business

Católica Lisbon School of Business and Economics, Portugal · 2023 to Sep 2026

- Business statistics, financial decisions, derivatives. Thesis: identity verification without a single trusted authority, tested with a 5,000-round simulation in Python.

Programmes and certifications

- Deep Dive into Blockchain, University of Zurich International Summer School, Jul 2026.
- Google for Startups Immersion EMEA, powered by Antler (accepted, Oct 2026); Solana Fall School (in progress).
- Hedera Business Foundation, Hedera Certified Foundation and Hashgraph Developer Course (2026); Introduction to Agents, Mercor (2026).

ENGINEERING SKILLS

- **Analysis and design:** ETABS, SAFE, Abaqus (finite element analysis), structural dynamics and modal analysis.
- **Codes:** ACI 318-19 reinforced concrete design and detailing; ASCE 7-16 loads; reading drawings against specifications.
- **Construction:** quantity takeoff and cost estimating, rebar and concrete quantities, site supervision of foundations and reinforcement.
- **Tools:** AutoCAD, Excel, Python.

RESEARCH

Financial Damage Detection Index

Sole author · Oct 2025 to Jan 2026

- Took the stiffness-based damage detection used in structural health monitoring and applied it to company finances, to catch financial stress before large price moves.
- Accepted at the IFABS 2026 London Conference (review score 7.4/10).

AWARDS

- Partner prize, ETHGlobal Lisbon 2026 (Autonomous On-Chain Automation Platform), for F2F Cross-Border.
- IFABS Grant for Young Researchers, IFABS 2026 London Conference.
- Full scholarship of 1,000 CHF, University of Zurich International Summer School, 2026.

OTHER EXPERIENCE AND PROJECTS

Generalist Expert, AI output evaluation

Mercor · Remote, freelance · Aug 2026

- Reviewed and rated AI answers for accuracy, reasoning, instruction-following and usefulness, and wrote down why.

Tnega: AI review of structural drawings, and Tnega Finance

Founder · 2026 · tnega.app · tnega.app/finance

- tnega.app: building an AI agent that will check reinforced concrete drawings against ACI 318-19 and ASCE 7-16 and send every life-safety finding to an engineer (in development).
- The site shows an interactive 3D code town of six concrete structure types (office building, residential tower, parking structure, foundations and retaining walls, bridge, mixed-use podium building). Each member shows its governing code provisions and where general AI models tend to go wrong.
- Tnega Finance (tnega.app/finance): shows the full cost of buying a tokenized stock or ETF on each chain, with a free MCP server that lets AI assistants compare costs and prepare orders the user signs in their own wallet.
- Wrote an escrow contract deployed on four chains; it passed 33 hostile test cases on a copy of the real network.

F2F Cross-Border and TrustCycle

Jul 2026 and 2026 · f2f-uzh.vercel.app · trustcycle.tech

- F2F: led the design of a cross-border transfer system where every compliance and routing decision is recorded on a public ledger.
- TrustCycle: decentralised academic recommendation letters; ranked 22 of 66 at the MasterZ and IOTA Web3 hackathon.

LANGUAGES

English (professional) · Portuguese (basic) · Persian (native)